ANDRÉ RAMOS

Junior Software Engineer (Backend & Cloud)

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SUMMARY

Recent Informatics Engineering graduate (BSc, Polytechnic Institute of Castelo Branco, June 2026) with hands-on project experience building Python backends, working with SQL and NoSQL databases, and deploying containerized applications on Microsoft Azure. Also built deep learning models that predict satellite collision risk. Looking for a junior software engineering role where I can learn from experienced engineers.

PROJECTS

PlantSnap – Cloud web application on Azure (university Cloud Computing course)
github.com/Andalexskywalker/plantsnap | Python, FastAPI, Docker, Azure, Cosmos DB (NoSQL), Terraform

- Built a REST API with FastAPI for plant identification, saving plants to a personal garden and user preferences, backed by Azure Cosmos DB (NoSQL) and Blob Storage, with Google login through App Service Authentication.
- Containerized the backend with Docker and deployed it on Azure App Service; added an Azure Functions timer that emails watering reminders and integrated third-party APIs (Azure Computer Vision, Plant.id).
- Defined the infrastructure as code with Terraform. Later added CI, Kubernetes manifests and monitoring with AI assistance; not deployed, as the course's Azure subscription has ended.

UAV Ground Control Station lab – simulated drone over MAVLink (personal project)
github.com/Andalexskywalker/uav-gcs-lab | ArduPilot SITL, QGroundControl, MAVLink/UDP, WSL2, Linux, WireGuard, ufw

- Set up a simulated drone (ArduPilot SITL on WSL2) and a ground control station on a second Linux machine, connected over MAVLink/UDP, and documented the problems I ran into and how I fixed them.
- Restricted the link with ufw firewall rules, tunnelled it through WireGuard, checked it with tcpdump, measured the latency added by the tunnel (about 0.8 ms) and tested how the link recovers from a 20-second outage.

SMCTE – Satellite collision risk prediction (final BSc project, team of 2)
github.com/Andalexskywalker/SMCTE | Python, pandas, scikit-learn, TensorFlow/Keras, Streamlit

- Co-developed a pipeline on the ESA Collision Avoidance Challenge dataset (162,634 records across 13,154 events), with leakage-free event-level splits and training-only preprocessing.
- Compared six sequence models (LSTM, GRU, BiLSTM, BiGRU, 1D-CNN, TCN) on a highly imbalanced target (15% positive events); the best reached PR-AUC 0.29, about twice the 0.15 baseline of a random classifier. Built a Streamlit app for inference.

Friday – LLM desktop assistant (Python)
github.com/Andalexskywalker/Friday | Python, Google Gemini API, SQLite

- Built a Python assistant in which a Gemini model triggers local tools: a tool registry that maps the model's JSON calls onto functions, API model fallback, and retries with exponential backoff on rate limits.

Chess & Checkers – Java desktop game (university project)
github.com/Andalexskywalker/XadrezDamas | Java, Swing, object-oriented design

- Implemented the checkers and chess game logic (piece classes, move rules, mandatory capture, check and checkmate, promotion) in Java Swing, on top of a course-provided board GUI.

SKILLS

Programming: Python (FastAPI, pandas, scikit-learn, TensorFlow/Keras), Java (university project), Bash, REST APIs
Databases: SQL, SQLite, Azure Cosmos DB (NoSQL)
Cloud, Linux and tooling: Microsoft Azure (App Service, Functions, Blob Storage), Docker, Terraform, Git; Linux (WSL2), WireGuard, ufw, tcpdump
Currently learning: Kubernetes, CI/CD with GitHub Actions, Prometheus, Grafana

EDUCATION

BSc in Informatics Engineering – Polytechnic Institute of Castelo Branco (IPCB)
Graduated June 2026 (degree completed). Relevant coursework: Internet of Things, Big Data, Cloud Computing, Cybersecurity.

LANGUAGES

Portuguese (native); English (C1, EF SET 65/100, 2025)